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Relativistic Transformations of Thermodynamics, Relativistic Statistical Mechanics and Einstein's Dual Theory

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Abstract. A brief description of the Redefined Relativistic Thermodynamics is exposed relating it with the Relativistic Statistical Mechanics and showing that Einstein-Planck, Ott and Rohrlich proposals represent particular choices of a reference frame where the instantaneity is considered. Einstein's dual theory described arriving at the conclusion that for a particle system, there is a universal time called the proper time. The instantaneity can be considered in the frame in which the observer is at rest in the canonical dual Hamiltonian center of mass. This will relate the different proposals to the Proper Time of the system.

1 Introduction

In 1907, the laws of relativistic transformations of thermodynamics were proposed by Planck [1] and [2] (PE). There were no important discrepancies to the respect and all the theory was resumed in the books of Tolman [3] and Pauli [4]. However, at the end of his life Einstein communicated to van Laue [5] his doubts about the validity of the theory. From it, in 1964, Ott [6] published his posthumous paper where he proposed a new set of relativistic transformation laws in thermodynamics and the subject newly called the attention of many researches (Ott proposal (O)). Consequently, countless number of different proposals were published [7]. More than a hundred of articles appeared during the sixties and the seventies of the past century defending or contradicting the PE proposal. Many theories were presented incorporating new variants. For example, we can mention the works realised by Arzeliès [8], Rohrlich [9] and Landsberg [10] (Landsberg proposal (L)) among others. Landsberg [11] suggested that just the experiment could clarify the controversies. Nevertheless, in 1968, Balescu [7] presented a statistical theory of the PE proposal demonstrating that not only it was the unique model which preserves the invariant form of thermodynamics but any other scheme could be obtained by using a gauge transformation. However, at the beginning of this century, most of the papers were close to the PE theory as it is the case of Sieniutycz's article [12]. In the mean time, Landsberg and Matsas [13], [14] published two works where they claimed having demonstrated that due to the equivalence between the frequency number density for a moving black-body and the excitation rate of the Unruh-DeWitt detector [15], a relativistic transformation law of the temperature cannot be defined. Their reasoning was based on the non-Planckian form of the frequency number density. Apparently, the Landsberg and Matsas papers explained the origin of all controversies. However, Ares de Parga et al [16] (the AA proposal) integrated the frequency density



number over all the particle frequencies and obtained the real number of particles. Then they calculated the internal energy multiplying the frequency number density by the frequency quantum of energy and integrating over all the frequencies, they got the PE transformation law of the energy. Using finite time thermodynamics, Ares de Parga et al [16] found an exact differential which permitted them to define thermal quantities which preserve the invariant form of thermodynamics that was misconceived in the PE and Balescu proposals. The AA proposal was based on comparing the transformation law of the free Helmholtz energy [17] with the transformation law of the internal energy. The final conclusion consists in accepting that the redefined relativistic thermodynamics permits not only a well definition of the temperature and its relativistic transformation but also an invariant form of the Thermodynamics. In this order of ideas, we can assure the existence of a relativistic temperature, which transforms as: $T = \gamma^{-1}T_0$, where T , T_0 are the temperatures at the moving system and at the rest frame respectively and $\gamma = 1/\sqrt{1-u^2/c^2}$, where u represents the velocity between both systems. With this result, we are able to analyze the black-body distribution of $2.7K$ radiation background [18].

The corrected theory will be called the Redefined Relativistic Thermodynamics [RRT] [19] and it needs to take into account the Nakamura concept of the volume [20] which permits us to define the energy momentum of a system in a covariant form. If a particular choice of the volume is made, one of the different Rohrlich proposals [9] of the relativistic transformation law of the heat appears as a consistent theory. Indeed, three different consistent models can be deduced from this theory and they are described by considering the instantaneity in a system and the definition of the rest frame of the volume. By using two blackbodies with different velocities, it can be shown that the van Kampen theory [21] do not conserve the heat and for this reason it must be discarded. By using the Henry temperature [22], [23] and the excitation rate of the Unruh DeWitt detector [22], [23], [24] and using the RRT, a relativistic thermometer can be proposed.

Nevertheless, by using a different method, Lineweaver [25] has shown that the direction of the reference frame such that the black-body which carries on the $2.7K$ radiation background is at rest, corresponds to the Hydra-Centaur direction with a speed of $627km/sec$. Therefore, since the speed of the reference frame is $627km/sec$, which is very small in comparison with the speed of the light, we can assure that the temperature $2.7K$ is correct; that is: it represents the temperature of the CMB at rest since $\gamma \simeq 1$. If the detector is put in the Hydra-Centaur direction with a speed of $627km/sec$, the particle number density will have to be independent of the angle, that is, the dipolar effect should disappear. Einstein's centennial commemorative congress at the University of Illinois at Carbondale, Dirac [26] claimed that due to the dipole anisotropy of the radiation background, it will be possible to find the velocity of such a frame. By extrapolation, we will be able to determine the reference frame where the Big-Bang occurred and it will represent a privileged frame contradicting the general theory of relativity. In the same congress, differing from Dirac's argument, Wigner [26] commented that general relativity just assures the equivalence of the laws of the nature in each reference frame and it indicates nothing about the initial conditions and so the discovery of a reference frame at rest with the Big-Bang will not contradict the general theory of relativity. The second comment corresponds to note that Henry et al [23] by the time when the $2.7K$ radiation background was discovered, assured that the sole effect of a uniform motion through black-body cavity is to introduce an effective temperature which replaces the rest frame cavity temperature for each angle. If not, Wigner will be wrong and Dirac will possess the truth [26].

The article is organized as follows: in Section 2, the Redefined Relativistic Thermodynamics is presented giving the new definitions of the different quantities and obtaining the three principal proposals; namely: The PE proposal, the O proposal and the L proposal. In section 3, the Statistical Mechanics is described by its corresponding relativistic transformation laws. In Section 4, a brief description of the Einstein's Dual Theory is exposed including the role that plays by the proper time. In Section 5, the RRT is related to Einstein's Dual Theory, concluding that the ROG, the PE and the L proposals are compatible with the Einstein's Dual Theory. Some concluding remarks are given in Section 6.

2 Relativistic Transformation Laws of Thermodynamic Quantities

From the beginning of Special Relativity, Fermi noticed that the sum of vectorial quantities along a 4-volume or 3-hypersurface is not a well-defined quantity unless some divergences vanish [39], [40], [41], [9], [42], [43]. This result obligates to well define the quantities that we will integrate, as the 4-volume or the 3-hypersurface, etc. Therefore, first of all, let's give a brief overview of Redefined Relativistic Thermodynamics [RRT]. Mathematically, as Nakamura [20] proposed, we can describe the volume as follows: A three dimensional flat plane is defined as a set of events that satisfies

$$\omega^\mu x_\mu = 0, \quad (1)$$

where ω^μ represents the time-like unit constant vector that defines the direction of the three-dimensional instantaneous space in the four-dimensional space. The volume $V_K(w)$ is defined as the intersection of this flat plane and the world tube of the object. If we consider a frame K and the volume V_{rest} in its rest frame K_{rest} , the unit vector $\mathbf{u}$ represents the motion between K and K_{rest} . The volume is defined as

$$V(w) = \frac{V_{rest}}{u^\mu \omega_\mu}, \quad (2)$$

and the corresponding 4-vector volume as

$$V^\mu = \frac{\omega^\mu V_{rest}}{u^\nu \omega_\nu}. \quad (3)$$

2.1 General Thermodynamics in Minkowski Space

In general, we define

$$\xi^\mu = \frac{\omega^\mu}{u^\lambda \omega_\lambda} E_{rest}, \quad (4)$$

and

$$d\Theta^\mu = \frac{\omega^\mu}{u^\lambda \omega_\lambda} dQ_{rest}, \quad (5)$$

where E_{rest} and Q_{rest} represent the energy and the heat, both in the reference frame at rest. From Eqs. (3), (4), (5), we propose a generalization of the first law of thermodynamics as

$$d\xi^\mu = d\Theta^\mu - PdV^\mu = d\Theta^\mu - dW^\mu, \quad (6)$$

with

$$d\xi^\mu = \omega^\mu \frac{dE_{rest}}{u^\lambda \omega_\lambda}, \quad d\Theta^\mu = \omega^\mu \frac{dQ_{rest}}{u^\lambda \omega_\lambda} \quad \text{and} \quad dW^\mu = \omega^\mu \frac{PdV_{rest}}{u^\lambda \omega_\lambda}, \quad (7)$$

P is the invariant pressure. Define the 4-vector temperature as

$$T_{PE}^\mu = \frac{\omega^\mu}{u^\lambda \omega_\lambda} T_{rest}, \quad (8)$$

The question now is how to express covariant relativistic thermodynamics independent of the choice of the volume; Also, define $e_{rest} = E_{rest}/V_{rest}$, a 4-vector G^μ by

$$\begin{aligned} G^\mu &= [(P + e_{rest})u^\mu u^\nu - (P + e_{rest})g^{\mu\nu}] \frac{V_{rest}\omega_\nu}{u^\lambda \omega_\lambda} \\ &= (PV_{rest} + E_{rest})u^\mu - (PV_{rest} + E_{rest}) \frac{\omega^\mu}{u^\lambda \omega_\lambda} \end{aligned} \quad (9)$$

and P^μ as

$$P^\mu = (PV_{rest} + E_{rest})u^\mu - PV_{rest} \frac{\omega^\mu}{u^\lambda \omega_\lambda}, \quad (10)$$

and then we construct the momentum-like 4-vector:

$$\begin{aligned} &P^\mu - G^\mu \\ &= (PV_{rest} + E_{rest})u^\mu - PV_{rest} \frac{\omega^\mu}{u^\lambda \omega_\lambda} \\ &\quad - (PV_{rest} + E_{rest})u^\mu + (PV_{rest} + E_{rest}) \frac{\omega^\mu}{u^\lambda \omega_\lambda} \\ &= \frac{\omega^\mu}{u^\lambda \omega_\lambda} E_{rest} = \xi^\mu. \end{aligned} \quad (11)$$

We have demonstrated that ξ^μ can be obtained from different definitions which are equivalent, Eqs. (4) and (11). Therefore, we can assure that for any chosen volume, a covariant relativistic thermodynamics

that we will call the Redefined Relativistic Thermodynamics [RRT], can be described by the following relations

$$d\xi^\mu = d\Theta^\mu - PdW^\mu, \quad (12)$$

where

$$d\xi^\mu = \omega^\mu \frac{dE_{rest}}{u^\lambda \omega_\lambda}, \quad d\Theta^\mu = \omega^\mu \frac{dQ_{rest}}{u^\lambda \omega_\lambda} \quad \text{and} \quad dW^\mu = \omega^\mu \frac{PdV_{rest}}{u^\lambda \omega_\lambda}, \quad (13)$$

$$T^\mu = \omega^\mu \frac{T_{rest}}{u^\lambda \omega_\lambda} \quad \text{and} \quad \beta_\mu = \frac{u_\mu}{kT_{rest}}, \quad (14)$$

and with the entropy expressed as

$$dS = \beta_\mu \frac{V_{rest} \omega_\mu}{u^\lambda \omega_\lambda} dT^{\mu\nu} + \beta_\mu PdV^\mu, \quad (15)$$

where the energy-momentum density tensor is defined by

$$T^{\mu\nu} = (P + e_{rest})u^\mu u^\nu - Pg^{\mu\nu}, \quad (16)$$

2.2 Case 1 or Rohrlich-Ott-Gamba-like proposal [ROG]:

If we want to deal with the volume that represents all the events that simultaneously occur in the rest frame of the volume, but if we look at this volume from the frame K (the measuring devices), we have the following

$$\omega^\mu = \left(\gamma, \gamma \frac{\mathbf{u}}{c} \right) \quad \text{and} \quad u^\mu = \left(\gamma, \gamma \frac{\mathbf{u}}{c} \right). \quad (17)$$

Therefore,

$$u^\mu \omega_\mu = 1. \quad (18)$$

The volume considered is

$$V_{K_{rest}} = \frac{V_{rest}}{u^\mu \omega_\mu} = V_{rest}, \quad (19)$$

and the corresponding 4-vector volume is represented by

$$V_{K_{rest}}^\mu = \frac{\omega^\mu V_{rest}}{u^\nu \omega_\nu} = \omega^\mu V_{rest} = u^\mu V_{rest} = \left(\gamma, \gamma \frac{\mathbf{u}}{c} \right) V_{rest}. \quad (20)$$

Therefore, in this case, the differential of the volume is:

$$dV_{K_{rest}}^\mu = \frac{u^\mu dV_{rest}}{u^\nu \omega_\nu} = \omega^\mu dV_{rest} = u^\mu dV_{rest} = \left(\gamma, \gamma \frac{\mathbf{u}}{c} \right) dV_{rest}. \quad (21)$$

which represents the 4-vector differential of the volume that Gamba [43] has used in order to clarify the electromagnetic controversy of the 4/3 term.

If we notice that in the ROG case, G^μ vanishes, that is,

$$G^\mu = (PV_{rest} + E_{rest})u^\mu - (PV_{rest} + E_{rest})u^\mu = 0, \quad (22)$$

we have $P_{ROG}^\mu = \xi_{ROG}^\mu$.

2.2.1 Thermodynamics of the Rohrlich-Ott-Gamba-like proposal The thermodynamical relations are obtained just by defining the volume. The Ott-Rohrlich proposal is obtained by choosing the RG volume. A corrected redefined relativistic Thermodynamics is also found by using the PE volume. The interesting fact is that for any other choice of the volume we will obtain the thermodynamical relations. It can be thought that this new theory does not contribute to thermodynamics, since we have relations of the form $\gamma dE_{rest} = \gamma dQ_{rest} - \gamma PdV_{rest}$ which can be simplified to $dE_{rest} = dQ_{rest} - PdV_{rest}$, that is, the thermodynamics in the rest frame. However, when two systems in different frames are considered to interact, the above relation is not longer valid and it has to be modified. This will be applied to the study of two black-bodies that interact with the same initial conditions but with different motions.

Before finishing this section, it has to be pointed out that the validity of the theory must satisfy some constraints. The idea can be summarized by the following sentences due to Nakamura [20]: "This means

the change is slow enough to be able to regard objects in interest is in equilibrium all the time. There can be another requirement in relativistic thermodynamics; the change must be much slower than the transit time scale of the light across the object. However, this requirement is satisfied when we can regard the process as adiabatic, since an object cannot be in equilibrium within a time scale shorter than the transit time of light.” A summary of the ROG is represented in Table 1.

V^0	P	S	T^0	Θ^0	ξ^0	F^0
γV_o	P_o	S_o	γT_o	γQ_o	γE_o	γF_o

Table 1

2.3 Case 2 or Planck-Einstein-like proposal [PE]

Let us now consider a volume generated by a Lorentz contraction in K of a body whose volume V_{rest} is at rest in K_{rest} . This volume is equal to

$$V_K = \gamma^{-1} V_{rest}. \quad (23)$$

in K . As we noticed before, all the points of this volume are simultaneous in K and, consequently, we can consider it as a volume at rest in K . In the Nakamura formalism, the different 4-vectors will be described in K by

$$\omega^\mu = (1, \mathbf{0}) \quad \text{and} \quad u^\mu = \left(\gamma, \gamma \frac{\mathbf{u}}{c} \right). \quad (24)$$

Therefore,

$$u^\mu \omega_\mu = \gamma. \quad (25)$$

The volume considered is

$$V_{K_{rest}} = \frac{V_{rest}}{u^\mu \omega_\mu} = \frac{V_{rest}}{\gamma} = \gamma^{-1} V_{rest}, \quad (26)$$

and the corresponding 4-vector volume is represented by

$$V_{K_{rest}}^\mu = \frac{\omega^\mu V_{rest}}{u^\nu \omega_\nu} = \omega^\mu \frac{V_{rest}}{\gamma} = (1, 0) \frac{V_{rest}}{\gamma}. \quad (27)$$

Therefore, in this case, the differential of the volume is as follows:

$$dV_{K_{rest}}^\mu = \frac{\omega^\mu dV_{rest}}{u^\nu \omega_\nu} = \omega^\mu \frac{dV_{rest}}{\gamma} = (1, 0) \frac{dV_{rest}}{\gamma}. \quad (28)$$

2.3.1 Thermodynamics of the Planck-Einstein-like proposal Let us present a special resume of the theory when the PE volume is chosen in a not covariant form. Indeed, the last formalism keeps the invariance of the form in thermodynamics by defining the following quantities: the volume $V = V_{PE}^0 = \gamma^{-1} V_{rest}$, the redefined energy ξ

$$\xi = E - \gamma(E_{rest} + PV_{rest}) \frac{u^2}{c^2} = \xi^0 = \gamma^{-1} E_{rest}, \quad (29)$$

where E represents the relativistic PE transformed internal energy. It has to noticed that ξ contains the Einstein bulk energy described by

$$W = \gamma(E_{rest} + PV_{rest}) \frac{u^2}{c^2}. \quad (30)$$

Table 2 describes the more important transformed quantities: (F represents the Helmholtz free energy)

V^0	P	S	T^0	$d\Theta^0$	ξ^0	F^0
$\frac{V_{rest}}{\gamma}$	P_{rest}	S_{rest}	$\frac{T_{rest}}{\gamma}$	$\frac{dQ_{rest}}{\gamma}$	$\frac{\xi_{rest}}{\gamma} = \frac{E_{rest}}{\gamma}$	$\frac{F_{rest}}{\gamma}$

Table 2

The difference with the “renormalized” relativistic thermodynamics consists of noticing that the heat has been redefined as the redefined heat and the work has kept its form without adding to it the bulk energy. Indeed, the work transforms as $dW = P\gamma^{-1}dV_{rest}$, where P represents the pressure. With this definition, we obtain all the thermodynamical relations, as for example, the first law, that is,

$$d\xi = d\Theta - PdV. \quad (31)$$

2.4 Landsberg-like Proposal [L]

If we want to measure at K and the rest volume is in K

$$\omega^\mu = (1, \mathbf{0}) \quad \text{and} \quad u^\mu = (1, \mathbf{0}) \quad (32)$$

$$u^\mu \omega_\mu = 1, \quad V = \frac{V_{rest}}{u^\mu \omega_\mu} = V_{rest}, \quad V^\mu = \frac{w^\mu}{u^\mu \omega_\mu} V_{rest} = (1, 0) V_{rest}, \quad (33)$$

$$dV^\mu = \frac{\omega^\mu}{u^\mu \omega_\mu} dV_{rest} = (1, 0) dV_{rest}. \quad (34)$$

Table 3 describes the most important transformed quantities: (F represents the Helmholtz free energy).

V^0	P	S	T^0	$d\Theta^0$	ξ^0	F^0
V_{rest}	P_{rest}	S_{rest}	T_{rest}	dQ_{rest}	$\xi_{rest} = E_{rest}$	F_{rest}

Table 3

Of course this L proposal just consider that we are measuring all the thermodynamics quantities in the frame where the thermodynamic system (the volume) is at rest. Many authors consider that this is the way we have to deal with thermodynamic system, however when we are dealing with a system composed of a Mixing of gases with different relative velocities [44], it is necessary to use the ROG or the PE proposals, or to deal with the center of mass of the whole system as we will see later.

3 Statistical Mechanics and the Relativistic Transformation Laws

On the other hand, it is necessary to deal with Relativistic Statistical Mechanics and Canonical transformation. Curie et al. [27] and later Balescu and Kotera [28] have demonstrated that the distribution function in a moving system, with velocity u in the x -axis with respect to the rest frame of the system, can be expressed as [30], [31]:

$$f(q, p; s) = e^{[K_1]s} f(q, p, 0) \quad (35)$$

where q and p represent the canonical variables, $u = \tanh s$ and K_1 is the boost generator in the x -axis. Balescu [28], as we noticed above, partially used this result but he directly did not calculate the distribution function. It is a well known result that a canonical transformation form a a group. Canonical transformations are analytic functions of continuous parameters of separate groups belonging to the class known as Lie groups. In order to solve the last equation, let us firstly expose some important results. The exponential of the boost generator applied to a function A , can be expressed as:

$$e^{[K_1]s} A = A + s[A, K_1] + \frac{1}{2!} s^2 [[A, K_1], K_1] + \dots, \quad (36)$$

which has been demonstrated by Balescu and Kotera [28] and used by Balescu in his famous article entitled Relativistic Statistical Thermodynamics [7]. Indeed, let us notice the regular Poisson brackets expressions (no quantum case); that is:

$$\begin{aligned} [H, P_1] &= 0 \\ [K_1, H] &= P_1 \\ [K_1, P_1] &= H \end{aligned} \quad (37)$$

where H and P_1 represent the Hamiltonian and the total momentum in the x -axis of the system, respectively. Then, Balescu, starting from

$$f(q, p, 0) = e^{\frac{1}{kT_0} [F(T_0, V_0, 0) - H(q, p)]}, \quad (38)$$

where $[F(T_0, V_0, 0)]$ represents the Helmholtz free energy in the rest frame V_0 at temperature T_0 , arrive at

$$f(q, p, s) = e^{[K_1]s} e^{\frac{1}{kT_0} [F(T_0, V_0, 0) - H(q, p)]}. \quad (39)$$

Finally,

$$f(q, p, s) = e^{\frac{1}{kT} [F - \Pi(q, p)]}, \quad (40)$$

with

$$T = \gamma^{-1} T_0, \quad F = \gamma^{-1} F_0 \quad \text{and} \quad \Pi = H - P_1 \tanh s. \quad (41)$$

That is: the canonical distribution function, which is a probability function of the generalized canonical coordinates q and p , has been derived [32] and it is expressed by:

$$\begin{aligned} f(q, p, s) &= \exp \frac{1}{kT} [F - \Pi] \\ &= \exp \frac{\gamma}{kT_{rest}} [\gamma^{-1} F_{rest} - \gamma^{-1} H(q, p)] = f(q, p, 0), \end{aligned} \quad (42)$$

where $\Pi = H - P_1 c \tanh s$ represents the redefined Hamiltonian and $\tanh s = \frac{u}{c}$. It has to be noticed that this representation of Statistical Mechanics is consistent with the PE proposal. This is because the probability function is defined by using the instantaneity and the volume as it is proposed in the PE proposal.

We can write the partition function as [32]

$$\begin{aligned} Z(s, T) &= \frac{1}{N! h^{3N}} \int d^{3N} p \exp \left[-\frac{1}{kT} \Pi \right] \\ &= \frac{1}{N! h^{3N}} \int d^{3N} p_0 \exp \left[-\frac{\gamma}{kT_0} \frac{H(q_0, p_0)}{\gamma} \right] \\ &= Z(0, T_0) \end{aligned} \quad (43)$$

which is an invariant. Using the Campbell–Hausdorff relation [29], López Carrera and Ares de Parga have demonstrated [32] that in Quantum Mechanics, the quantum partition function can be taken as

$$Q(s) = \text{Tr} \exp \left[\frac{1}{kT} \Pi \right] = \sum_i \beta' E'_i \quad (44)$$

with $\beta' = \gamma/kT$ and $E'_i = E_i - uG_x = \xi_i$, where G_x corresponds to

$$\mathbf{G} = \frac{4}{3} \gamma E_{i0} \frac{\mathbf{u}}{c^2}. \quad (45)$$

$\mathbf{G}$ corresponds to G^μ for the PE proposal.

The distribution function is [33]

$$f = \frac{1}{Z} e^{-\beta_\mu \Pi^\mu} \quad (46)$$

with

$$\beta_\mu \Pi^\mu = \frac{1}{kT} \sum_{i=1}^n \sqrt{p_i^2 c^2 + m^2 c^4} - uP. \quad (47)$$

Arrive at

$$f(\mathbf{v}) = \frac{m^3 \gamma^5(\mathbf{v})}{\gamma(\mathbf{u}) \Upsilon_3(\gamma(\mathbf{u}) T)} \exp -\frac{m \gamma(\mathbf{u}) \gamma(\mathbf{v})}{kT} (1 - \mathbf{v} \cdot \mathbf{u}) \quad (48)$$

where

$$\Upsilon_3(y) = \frac{4\pi}{h^3} (mc)^3 \frac{K_2\left(\frac{mc^2}{ky}\right)}{mc^2} ky, \quad (49)$$

with K_2 represents the Bessel function of second rank.

4 Einstein's Dual Theory and the Proper Time

We need to notice that in Einstein's Dual Theory [34], [35], [36], [37], [38] the proper time τ remains invariant, and we assume K invariant.

$$K = \frac{\pi^2}{2} + mc^2 + \frac{V^2}{2mc^3} + \frac{V\sqrt{c^2\pi^2 + m^2c^4}}{mc^2}, \quad H_0 = \sqrt{c^2\pi^2 + m^2c^4}, \quad (50)$$

$$\frac{d\mathbf{x}}{d\tau} = \frac{\partial K}{\partial \mathbf{p}} = \frac{H}{mc^2} \left(\frac{c^2\pi}{H_0} \right) = \frac{b}{c} \left(\frac{c^2\mathbf{x}}{H_0} \right) \Rightarrow \frac{d\mathbf{x}}{d\tau} = \frac{b}{c} \frac{d\mathbf{x}}{dt}, \quad (51)$$

$$\vec{\pi} = \mathbf{p} - \frac{e}{c} \mathbf{A} \quad (52)$$

One particle

$$\frac{d\mathbf{x}_i}{dt} = \mathbf{w}_i, \quad \frac{d\mathbf{x}_i}{d\tau_i} = \mathbf{u}_i, \quad \frac{d\mathbf{x}_i}{d\tau} = \mathbf{v}_i, \quad (53)$$

$$\text{where } b_i = \sqrt{u_i^2 + c^2} \quad \text{and} \quad b = \sqrt{U^2 + c^2} \quad (54)$$

$$\frac{\mathbf{w}_i}{c} = \frac{\mathbf{v}_i}{b} = \frac{\mathbf{u}_i}{b_i} \Rightarrow \gamma_i^{-1} = \sqrt{1 - \left(\frac{w_i}{c}\right)^2} = \sqrt{1 - \left(\frac{v_i}{b}\right)^2} = \sqrt{1 - \left(\frac{u_i}{b_i}\right)^2} \quad (55)$$

It should be noticed that eventually $\mathbf{U}$ may represent the motion of the canonical center of mass, as we shall see in the next section. The motion of the canonical center of mass is described by

$$\mathbf{X} = \frac{1}{H} \sum_i^N H_i \mathbf{x}_i + \frac{c^2 (\mathbf{S} \times \mathbf{P})}{H (Mc^2 + H)}, \quad (56)$$

where $\mathbf{S}$ is the global spin of the system of particles relative to O , the origin. Being

$$\mathbf{V} = \frac{d\mathbf{X}}{dt} \Rightarrow \mathbf{U} = \frac{d\mathbf{X}}{d\tau} = \gamma(\mathbf{V})\mathbf{V}. \quad (57)$$

The Hamiltonian H of many particles is

$$H = \sum_i H_i \quad \text{with} \quad H_i = H_{i0} + V_i = \sqrt{c^2\pi_i^2 + m_i^2c^4} + V_i \quad (58)$$

where

$$\vec{\pi}_i = \mathbf{p}_i - \frac{e_i}{c} \mathbf{A}_i \quad \text{with} \quad \mathbf{A}_i = \sum_{i \neq j} \mathbf{A}_{ij} \quad \text{and} \quad V_i = \sum_{i \neq j} V_{ij}. \quad (59)$$

Then

$$K = \frac{H^2}{2Mc^2} + Mc^2. \quad (60)$$

For Quantum Mechanics, we have

$$H_D = c\vec{\alpha} \cdot \vec{\pi} + mc^2\beta + V_0, \quad \text{with} \quad V = \frac{1}{2mc^2} [H_0 V_0 + V_0 H_0] \quad (61)$$

The Dirac Hamiltonian as:

$$K_D = \frac{H_D^2}{2mc^2} + \frac{mc^2}{2} = \frac{\pi^2}{2m} + V - \frac{e\mathbf{\Sigma} \cdot \mathbf{B}}{2mc} + mc^2 + \frac{V_0^2}{2mc^2}. \quad (62)$$

5 The Proper Time in Redefined Relativistic Thermodynamics

As we have seen in RRT, using the proper time theory, we just need to replace the value of

$$\mathbf{w} \Rightarrow \mathbf{V}, \quad \mathbf{u} \Rightarrow \mathbf{U} \quad \Rightarrow \frac{\mathbf{V}}{c} = \frac{\mathbf{U}}{b}. \quad (63)$$

Therefore,

$$\gamma^{-1} = \sqrt{1 - \left(\frac{\mathbf{V}}{c}\right)^2} = \sqrt{1 - \left(\frac{\mathbf{U}}{b}\right)^2} \quad (64)$$

$$b = \sqrt{U^2 + c^2} \Rightarrow \gamma^{-1} = \sqrt{1 - \left(\frac{\mathbf{U}^2}{U^2 + c^2}\right)} \quad (65)$$

$$\gamma^{-1} = \sqrt{\left(\frac{c^2}{U^2 + c^2}\right)} \Rightarrow \gamma = \sqrt{\frac{U^2 + c^2}{c^2}} = \frac{b}{c} \quad (66)$$

5.1 Ott-like proposal within the Einstein's dual theory

In this case, we have

$$\omega^\mu = \left(\gamma, \gamma \frac{\mathbf{V}}{c}\right) \Rightarrow \omega^\mu = \frac{b}{c} \left(1, \frac{\mathbf{U}}{b}\right) = \left(\gamma, \frac{\mathbf{U}}{c}\right) = \left(\frac{b}{c}, \frac{\mathbf{U}}{c}\right) \quad (67)$$

and

$$u^\mu = \left(\gamma, \gamma \frac{\mathbf{U}}{b}\right) = \left(\frac{b}{c}, \frac{\mathbf{U}}{c}\right) \quad (68)$$

Therefore,

$$u^\mu \omega_\mu = \frac{b^2}{c^2} - \frac{\mathbf{U}^2}{c^2} = \frac{b^2 - \mathbf{U}^2}{c^2} = \frac{c^2}{c^2} = 1. \quad (69)$$

The considered volume is

$$V_{K_{rest}} = \frac{V_{rest}}{u^\mu \omega_\mu} = V_{rest}, \quad (70)$$

and the corresponding 4-vector volume is represented by

$$V_{K_{rest}}^\mu = \frac{\omega^\mu V_{rest}}{u^\nu \omega_\nu} = \omega^\mu V_{rest} = u^\mu V_{rest} = \left(\gamma, \gamma \frac{\mathbf{U}}{b}\right) V_{rest} = \left(\frac{b}{c}, \frac{\mathbf{U}}{c}\right) V_{rest}. \quad (71)$$

5.2 PE-like proposal within the Einstein's dual theory

Let us now consider a volume generated by a Lorentz contraction in K of a body whose volume V_{rest} is at rest in K_{rest} . This volume is equal to

$$V_K = \gamma^{-1} V_{rest}. \quad (72)$$

in K . As we noticed before, all the points of this volume are simultaneous in K and, consequently, we can consider it as a volume at rest in K . In the Nakamura formalism, the different 4-vectors will be described in K by

$$\omega^\mu = (1, \mathbf{0}) \quad \text{and} \quad u^\mu = \left(\gamma, \gamma \frac{\mathbf{U}}{b}\right) = \left(\frac{b}{c}, \frac{\mathbf{U}}{c}\right). \quad (73)$$

Therefore,

$$u^\mu \omega_\mu = \gamma = \frac{b}{c}. \quad (74)$$

The volume considered is

$$V_{K_{rest}} = \frac{V_{rest}}{u^\mu \omega_\mu} = \frac{V_{rest}}{\gamma} = \gamma^{-1} V_{rest} = \frac{c}{b} V_{rest}, \quad (75)$$

and the corresponding 4-vector volume is represented by

$$V_{K_{rest}}^\mu = \frac{\omega^\mu V_{rest}}{u^\mu \omega_\mu} = \omega^\mu \frac{V_{rest}}{\gamma} = (1, 0) \frac{V_{rest}}{\gamma} = \left(\frac{c}{b}, 0\right) V_{rest}. \quad (76)$$

Therefore, in this case, the differential of the volume is as follows:

$$dV_{K_{rest}}^\mu = \frac{\omega^\mu dV_{rest}}{u^\mu \omega_\mu} = \omega^\mu \frac{dV_{rest}}{\gamma} = (1, 0) \frac{dV_{rest}}{\gamma} = \left(\frac{c}{b}, 0\right) dV_{rest}. \quad (77)$$

5.3 Landsberg Proposal

If we consider that we are thinking about defining the instantaneity in the rest frame of the system where $d\tau = dt$ and the volume $V = V_0$, we have

$$\omega^\mu = (1, \mathbf{0}) \quad \text{and} \quad u^\mu = (1, 0). \quad (78)$$

Therefore,

$$u^\mu \omega_\mu = 1. \quad (79)$$

The volume considered is

$$V_{K_{rest}} = \frac{V_{rest}}{1} = V_{rest}, \quad (80)$$

and the corresponding 4-vector volume is represented by

$$V_{K_{rest}}^\mu = (1, \mathbf{0}) V_{rest} = (1, \mathbf{0}) \frac{V_{rest}}{\gamma} = (1, 0) V_{rest}. \quad (81)$$

Therefore, in this case, the differential of the volume is as follows:

$$dV_{K_{rest}}^\mu = (1, \mathbf{0}) dV_{rest}. \quad (82)$$

We can define all the instantaneity in K_0 and the time is the same τ for all the system.

6 Concluding Remarks

The Proper Time in Redefined Relativistic Thermodynamics has been defined and therefore for each particle system if we can calculate or describe the motion of the canonical center of mass described by Eq. (52), we can physically describe the entire thermodynamics of a system in equilibrium. Many applications can be made from this result. Indeed, in the Introduction, we note that it can be understood from Redefined Relativistic Thermodynamics that the frame of reference such that the blackbody carrying the 2.7K background radiation is at rest corresponds to the Hydra-Centaur direction with a speed of 627 km/sec [18]. Also, the cases where two non-interacting gases with different temperatures collide can be easily solved by Redefined Relativistic Thermodynamics [44]. Other examples can be cited, but in all of them a measurement system has to be defined. However, Einstein's dual theory defines the center of mass of any interacting system that obtains equilibrium regardless of whether the particles interact or not. That is: in each proposal the rest frame of the system can be defined excluding the ideas of the Non-Interacting Theorem.

Therefore, the three proposals can be used to deal with a system of particles. First, if we want to deal with the volume that represents all the events that simultaneously occur in the rest frame of the volume but if we look at this volume from the frame K (the measuring devices), the ROG proposal is the right one. This method has been used by Gamba to clarify the electromagnetic controversy of the 4/3term [43], [32]. Second, when all the points of the volume are simultaneous in a reference frame K but the measuring devices are in a frame with velocity u with respect to the frame K , the PE proposal describes the thermodynamics of the system, even more so when it is a system composed of several gases that collide at different temperatures and reach equilibrium [44]. Third, the L proposal seems to be the most adequate technique to deal with a thermodynamic system which is at rest with the measuring devices. Therefore, for any system that can reach thermodynamic equilibrium, part of the problem is reduced to obtaining or calculating the canonical dual Hamiltonian center of mas of Einstein's Dual Theory.

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